

Ayaan Sanadi

Computer Vision & ML Engineer | Bengaluru, India

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CORE SKILLS

Programming Languages: Python, JavaScript, C, C++

Machine Learning : YOLO, MediaPipe, TensorFlow Lite (Micro), PyTorch

Frameworks & Tools: FastAPI, OpenCV, Next.js, Git

DataBases: PostgreSQL, MongoDB

PROFESSIONAL EXPERIENCE

Computer Vision Intern – Neuronetic Vision

Jan 2026 – Apr 2026

- Engineered end-to-end object detection pipelines by scraping, cleaning, and annotating complex image datasets to prepare high-quality training data for YOLO models.
- Trained, fine-tuned, and deployed YOLO models for real-time CCTV security systems, achieving robust detection of weapons (knives, handguns) and public spitting violations.
- Developed automated vision solutions for warehouse and facility management, creating models capable of identifying pests (rats) and detecting two-wheeler license plates in varied lighting conditions.

PROJECTS

Real-Time AI Smart Attendance & Analytics Pipeline

- Automated classroom attendance for 70+ students by engineering a production-ready computer vision pipeline utilizing YOLOv8-face and InceptionResnetV1 for live facial feature extraction.
- Maintained high FPS and prevented ID-switching by architecting a "Two-Brain" tracking system that paired short-term physics tracking (DeepOCSORT/Kalman filters) with long-term logical memory.
- Accelerated real-time video processing by offloading tensor operations to Apple Silicon (MPS backend), matching live frames to a vectorized master database via cosine similarity.
- Streamlined administrative reporting by integrating a Supabase (PostgreSQL) backend to translate AI outputs into official university registry data and dynamic CSV reports.

AI-Powered Acoustic Pest Detection System (Edge AI / TinyML)

- Detected agricultural pest infestations in real-time by deploying an INT8 post-training quantized CNN (under 65KB footprint) onto an ESP32 microcontroller using TFLite Micro.
- Replicated Python's Librosa library at the edge by building a custom C++ Digital Signal Processing (DSP) pipeline, converting raw 16kHz audio into Mel-Spectrograms on the hardware.
- Prevented micro-controller RAM overflow and hardware-level tensor corruption by manually forcing 16-byte memory alignment and dynamically shifting massive audio arrays to the Heap.
- Eliminated false-positive alerts by designing a 5-step rolling buffer with "3-out-of-5" voting logic, pushing verified HTTP POST alerts to a responsive Next.js dashboard.

FormCheck AI: Real-Time Biomechanics Analysis

- Built an AI-driven gym safety platform leveraging MediaPipe and OpenCV to perform real-time kinematic analysis and pose estimation for compound lifts.
- Delivered instant corrective audio cues and post-set analytics by implementing a custom validation pipeline that cross-referenced user form against domain-specific strength training metrics.

BirdSpec: End-to-End Species Detection Pipeline

- Classified 483 bird species by training a lightweight EfficientNet-V2 CNN on a dataset of 60,000+ images, handling high inter-class similarity and morphological differences.
- Deployed the inference engine via a FastAPI service and integrated it with a Next.js front-end and Supabase backend for user history tracking and Wikipedia context lookups.

EDUCATION

BMS Institute of Technology and Management

2024 – 2028

B.E. in Artificial Intelligence & Machine Learning | CGPA: 8.6